

How We Recovered the Digit, Part II

Project 2 method, taught slow

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This is a teaching guide for the method we use in Project 2. It is slower than the Project 1 lesson on purpose. The math is broken up into single-step moves. Every symbol gets named the moment it appears. Diagrams appear next to the math they describe.

By the end you should be able to say, in your own words, all of these:

1. What problem Project 2 asks you to solve.
2. What a diffusion model is and why it can act as a prior.
3. How the PnP-DM sampler turns the diffusion model into a tool for this problem.
4. Why we add a Gauss-Newton polish at the end.
5. Why the base polish over-fits, and how a TV-regularized polish fixes it.
6. How class-conditional diffusion with classifier-free guidance lets each chain commit to a digit class explicitly.
7. What the final number is and what it means.

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Part I — What are we solving?

1 Lesson 1: The problem in one paragraph

We have a square box with a hidden picture inside it. The picture is a handwritten digit, drawn as $28 \times 28 = 784$ pixels, each pixel a number that says how easily electricity flows at that point. We are not allowed to look inside. We can only push small electric currents through electrodes on the edge of the box and read voltages back, also at the edge. After many of these little experiments, we have a list of 1900 voltage numbers. *From only that list of 1900 numbers, we want to draw the digit that is inside.*

We will use the letter y for the list of 1900 readings, and the letter x for the $28 \times 28 = 784$ hidden pixels. We have a computer function F that, given any hidden picture x , tells us exactly which 1900 readings it would produce. So:

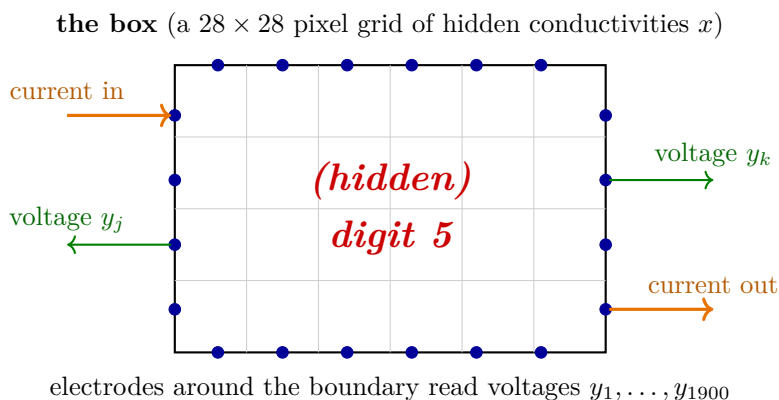
$$F(x) = y.$$

Going from x to y (the forward direction) is easy: just call F . Going from y back to x (the inverse direction) is hard, and that is our whole problem.

Project 2, in one line: given y_{obs} (1900 numbers), find the x (784 numbers) that produced it.

2 Lesson 2: A picture of the setup

Before any math, let's see what ERT actually is.



The box is a 28×28 grid. Each grid cell is one pixel of the hidden image x , which encodes “how easily electricity flows here.” The little blue dots around the edge are electrodes, and the orange arrows are one “little experiment” — we push current in one electrode and let it come

out another. While that current flows, we measure voltages at the other electrodes (green arrows). Over many such experiments, we collect 1900 voltage readings, which we stack into a column vector y .

Why does this even give us any information about the inside? Because electric current flows easily through high-conductivity material and struggles through low-conductivity material. The pattern of voltages at the edge depends on what’s inside. A blob of high conductivity in the middle “soaks up” current and produces a particular voltage pattern; move the blob, and the pattern shifts. So the readings carry information about the digit — just a garbled, edge-only, blurred-out version of it.

Analogy: It’s like a medical CT scanner that only has sensors on the outside of your body. The sensors don’t see the bones directly. They see how X-rays bent on their way through. From those bending patterns, software reconstructs an internal image. ERT is the electrical analog of that.

3 Lesson 3: Why this is hard

Two reasons.

Reason 1: many different x ’s give nearly the same y . The electrodes are far from the inside. A small feature inside the box barely shows up at the edge, and several different inside-pictures produce edge readings that differ only slightly. So if all we ask for is “find any x with $F(x) \approx y_{\text{obs}}$,” we get many candidates, including ugly noise-looking pictures that fit the readings to a few decimal places but look nothing like a digit.

Reason 2: tiny errors in y blow up in x . Even if the measurements are nearly perfect, the inverse map is so sensitive that small noise in y can cause wildly different reconstructions in x .

Both reasons say the same thing: pure data-fitting is not enough. We need an extra rule that says “the answer should also look like a real digit, not like noise.” That extra rule is called a *prior*, and choosing one is the whole game.

Analogy: If someone asks “what is $2+?=5$,” the answer is 3, unique. But if someone asks “what is $?+=5$,” the answer is any pair that sums to 5. We need a second rule, like “both numbers are between 0 and 5,” to narrow it down. The prior is that second rule.

4 Lesson 4: The rule for Project 2

Project 1 used a particular prior: “the answer is close to one of the 60,000 MNIST training digits.” That worked, but it is a very strong and very specific prior — essentially a lookup

table.

Project 2 says: **the prior must be a diffusion model**. A diffusion model is a neural network that has learned what a digit looks like, by practicing on lots of digits, and that can be used to generate fresh digit-shaped images at random. Once we have one, we have to combine it with our data y_{obs} to recover the actual digit.

This is the whole assignment. The rest of this guide is about how to take a diffusion model (which only knows about digits in the abstract) and use it to recover the specific digit that produced y_{obs} .

You can think of a diffusion model as a sketch artist who has practiced drawing thousands of digits. Ask them to sketch a digit at random, they can. Ask them “which digit produced these specific 1900 measurements?” and they have no idea — they have not seen the measurements. Our job is to combine the sketch artist with the measurements so they together produce the right digit.

Part II — What is a diffusion model?

5 Lesson 5: The strange but useful idea

A diffusion model is trained on one weird idea: take a clean picture, slowly destroy it with random noise until it is pure static, and train a neural network to *undo* that destruction one tiny step at a time.

Why does this teach the network anything useful? Because in order to undo the destruction one step at a time, the network has to learn what a slightly-less-noisy version of a digit picture looks like. After lots of practice on lots of training digits, it has effectively learned “what a digit looks like, in the presence of noise.” That turns out to be enough to do everything we want.

We will spend Lessons 6 through 13 building this up carefully. Take it slow. The math is not hard, but it has a lot of small moving parts, and each one shows up in later lessons.

6 Lesson 6: The forward chain, one step at a time

Pick a clean digit. We will call it x_0 . The subscript 0 means “zero noise, totally clean.” Think of x_0 as a 28×28 grid of numbers between 0 (black background) and 1 (bright stroke). Don’t worry about its exact pixel values; just hold the picture in your head.

We pick a tiny number β_1 . (The Greek letter “beta” is just a name for it.) In practice β_1 is something like 10^{-4} . We then build a slightly-noisier version of the picture, called x_1 :

$$x_1 = \sqrt{1 - \beta_1} x_0 + \sqrt{\beta_1} \varepsilon_1.$$

Let me unpack every piece of that line:

- ε_1 is a 28×28 grid of random numbers drawn from a standard Gaussian (bell curve, mean 0, variance 1). Each pixel of ε_1 is an independent random draw. So ε_1 on its own looks like pure TV-snow static.
- The first term, $\sqrt{1 - \beta_1} x_0$, is the clean picture shrunk by a tiny amount. Because β_1 is small, $\sqrt{1 - \beta_1}$ is very close to 1, so this is almost just x_0 itself.
- The second term, $\sqrt{\beta_1} \varepsilon_1$, is the static ε_1 scaled down by a small factor. Because β_1 is small, $\sqrt{\beta_1}$ is also small, so this contributes just a sprinkle of noise.
- Add the two together: we get x_1 , the original picture with a tiny bit of noise smeared on top.

So step 1 of the forward chain takes the clean picture and adds a sprinkle of noise. The picture still looks essentially identical to x_0 at this point.

Step 2 does the same thing again, with a slightly bigger β_2 :

$$x_2 = \sqrt{1 - \beta_2} x_1 + \sqrt{\beta_2} \varepsilon_2,$$

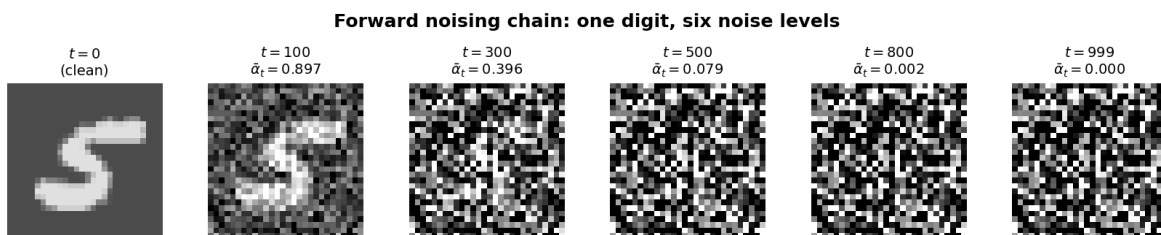
with a fresh random ε_2 . And then step 3, step 4, all the way up to step $T = 1000$. The β_t ’s grow as t grows, so the noise added per step gets bigger. By step 1000, the picture is *utterly gone* — x_{1000} is just pure Gaussian static, indistinguishable from what you’d get if you skipped the digit entirely.

Analogy: Imagine staring at a perfectly clear window. Every minute, someone breathes on it a little. After one breath, you can still see through. After 10 breaths, hazy. After 1000 breaths, it’s just fog and you can’t see anything. The forward chain is exactly that: the “view” is a digit and each “breath” is one step of noise.

The *forward chain* is a 1000-step process that turns a clean digit picture into pure noise. Each step adds a tiny sprinkle.

7 Lesson 7: A picture of the forward chain

Let’s actually look at what this destruction process looks like. Below is one MNIST digit (a 5) shown at six different points along the forward chain.



- At $t = 0$ (clean), it's just the digit.
- At $t = 100$, you can already see noise sprinkled on the background, but the 5 is still obvious.
- At $t = 300$, the noise has gotten substantial. You can *barely* make out the 5 if you squint.
- At $t = 500$, the 5 is mostly buried under noise. A neural network could still pick it out, but a human is having a hard time.
- At $t = 800$, almost nothing is left.
- At $t = 999$, just static. The original 5 is unrecoverable by eye.

The number $\bar{\alpha}_t$ under each panel (we'll define it formally in the next lesson) measures “how much of the original is left.” At $t = 0$, $\bar{\alpha}_t = 1$ (all of it). At $t = 999$, $\bar{\alpha}_t = 0.0005$ (essentially none of it).

This is what the forward chain looks like for one digit. For training, we run the chain on every training digit, getting a noisy version of every digit at every noise level. That's a lot of (noisy digit, noise level) pairs, and that's the training data for the network.

8 Lesson 8: A shortcut — the closed-form noising formula

We do not actually have to run all 1000 steps to know what x_t looks like. There is a one-line shortcut.

Define

$$\bar{\alpha}_t := \prod_{s=1}^t (1 - \beta_s).$$

The symbol $\prod$ just means “multiply these all together.” So $\bar{\alpha}_t$ is the product $(1 - \beta_1)(1 - \beta_2) \cdots (1 - \beta_t)$. It's the cumulative version of the per-step shrink factor.

Numerical example. Say β_t is constant at 10^{-4} for the first 100 steps. Then

$$\bar{\alpha}_{100} = (1 - 10^{-4})^{100} \approx 0.99.$$

After 100 steps, about 99% of the original signal remains. That matches what we see in Figure above: at $t = 100$, the digit is still very clearly visible.

After 1000 steps, with the schedule the model actually uses (growing β 's), $\bar{\alpha}_{1000}$ is around 5×10^{-4} . Almost no signal left.

The shortcut. Instead of running t small steps, you can write

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I).$$

That is: x_t is just x_0 shrunk by $\sqrt{\bar{\alpha}_t}$, plus *one* Gaussian noise sample scaled by $\sqrt{1 - \bar{\alpha}_t}$. No chain of 1000 steps required. (Why is this allowed? Because adding independent Gaussians at every step still gives you a Gaussian at the end, with a known variance.)

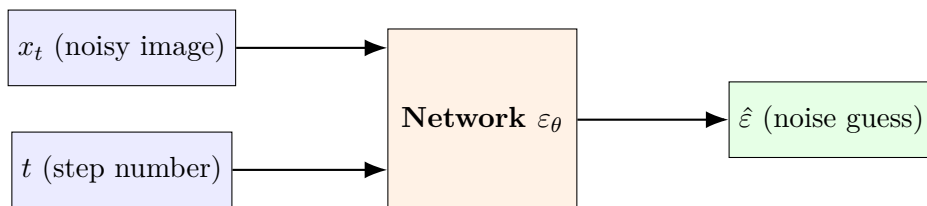
- When $t = 0$: $\bar{\alpha}_0 = 1$, so $x_0 = 1 \cdot x_0 + 0 \cdot \varepsilon = x_0$. Clean, as expected.
- When $t = T = 1000$: $\bar{\alpha}_T \approx 0$, so $x_T \approx 0 \cdot x_0 + 1 \cdot \varepsilon = \varepsilon$. Pure noise, as expected.
- In between, $\bar{\alpha}_t$ slides from 1 down to 0. So $\sqrt{\bar{\alpha}_t}$ slides from 1 down to 0 (the original fades out) and $\sqrt{1 - \bar{\alpha}_t}$ slides from 0 up to 1 (the noise takes over).

$\bar{\alpha}_t$ is the “how much of the original is still here” at step t . It is the single number that captures where we are in the forward chain. You will see this letter on almost every page.

9 Lesson 9: What the network does — inputs and outputs

The forward chain is just math. The reverse direction — recovering the digit from a noisy version — is what we train a neural network to do.

The contract. The network takes two inputs — a noisy picture and which step we are at — and produces one output — a guess at the noise that was added.



We write the network as $\varepsilon_\theta(x_t, t)$. The little θ subscript is just a reminder that the network has internal weights that we have to learn during training; once it’s trained you can ignore θ and just call it “the network.”

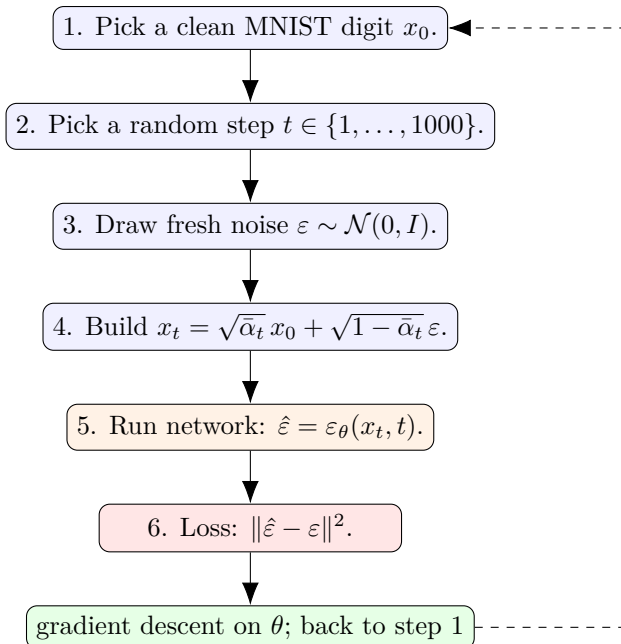
The output $\hat{\varepsilon}$ has the same shape as x_t (so 28×28). It’s the network’s guess at the noise sample that was used to build x_t from x_0 .

Why predict the noise rather than the clean image? Predicting noise turns out to be easier and more numerically stable in practice. But it doesn’t really matter — once you can predict the noise, you can also recover the clean image, with one line of algebra. (Lesson 11.)

What does the network look like inside? A so-called U-Net: some convolutional layers that progressively compress the image, then some that expand back to 28×28 . The architecture details don't matter for our purposes. What matters is the contract above.

10 Lesson 10: Training the network

We train the network so that for any clean picture, any step t , and any noise sample ε , it correctly predicts ε . The training loop has six steps:



After many passes through the training data, the network gets good at predicting noise from any (noisy-picture, step) pair.

Notice the cost of training data: zero. For each clean training digit x_0 we can synthesize 1000 training pairs (one per possible t). We never had to collect “ground-truth pairs” from somewhere; the forward chain manufactures them on demand.

The pretrained MNIST DDPM we use, `laurent/ddpm-mnist` on Hugging Face, has about 1 million internal weights and was trained this way on standard MNIST for a few hours on one GPU. We just downloaded it; we did not train it from scratch.

11 Lesson 11: Tweedie’s formula — recovering the clean image from the noise guess

OK here is where we cash in. The network predicts the *noise*. What we really want is a guess at the *clean image* that the noise was added to. The good news is: if we have one, algebra gives us the other.

Go back to the closed-form noising formula:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon.$$

This is a relationship between three things: x_t (which we see), x_0 (what we want), and ε (which the network can guess).

Pretend the network’s guess $\hat{\varepsilon}$ is exact, and *solve* the formula for x_0 . Step by step.

Step 1. Start with

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \hat{\varepsilon}.$$

Step 2. Subtract $\sqrt{1 - \bar{\alpha}_t} \hat{\varepsilon}$ from both sides to isolate the term containing x_0 :

$$x_t - \sqrt{1 - \bar{\alpha}_t} \hat{\varepsilon} = \sqrt{\bar{\alpha}_t} x_0.$$

Step 3. Divide both sides by $\sqrt{\bar{\alpha}_t}$ to solve for x_0 alone:

$$\frac{x_t - \sqrt{1 - \bar{\alpha}_t} \hat{\varepsilon}}{\sqrt{\bar{\alpha}_t}} = x_0.$$

That’s it. The thing on the left is the network’s implicit guess at the clean image, given x_t and the network’s noise prediction. We give it its own name $\hat{x}_0$:

$$\hat{x}_0(x_t, t) := \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \varepsilon_{\theta}(x_t, t)}{\sqrt{\bar{\alpha}_t}}.$$

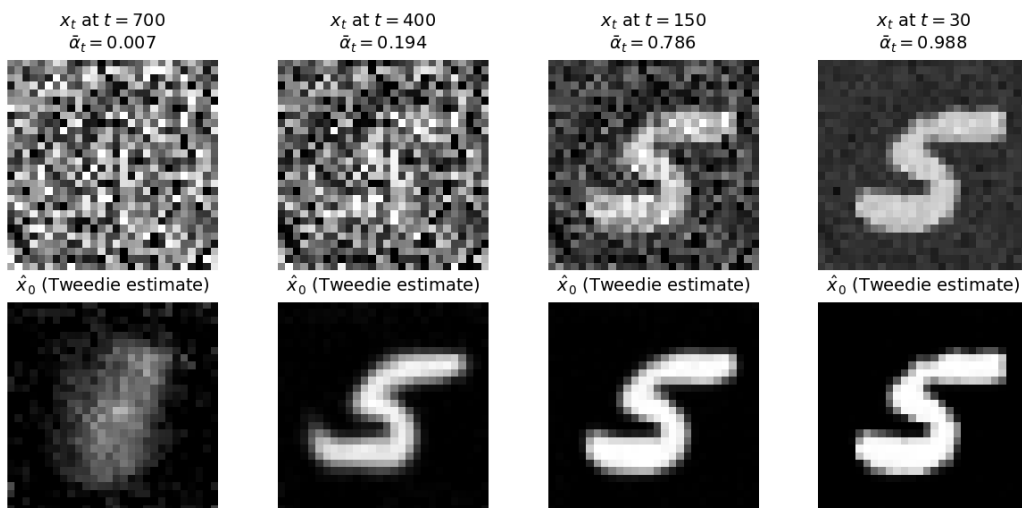
This is called **Tweedie’s formula** (after Maurice Tweedie). There is a deeper statistical reason why it’s exact in expectation, but for our purposes it is just “the noising formula solved for x_0 , using the network’s noise prediction as a stand-in for the unknown ε .”

$\hat{x}_0$ is the *denoiser as oracle*. At any moment along the noisy chain, ask the network for its $\hat{x}_0$ and you get a guess at the underlying clean digit. We will call this oracle constantly.

12 Lesson 12: A picture of Tweedie's estimates

The Tweedie formula is mathematical, but its output is a literal image we can look at. Below, the top row shows x_t at four different noise levels (the noisy inputs). The bottom row shows the corresponding $\hat{x}_0$ (the network's guess at the clean image).

Tweedie's formula at four noise levels (top: input x_t ; bottom: network's clean-image guess $\hat{x}_0$)



What you should see:

- At $t = 700$ (top-left): x_t is mostly noise (only a faint shape is visible). The network's $\hat{x}_0$ is a blurry but recognizable 5. The network has committed to “a 5 is in here somewhere” but the strokes are fuzzy.
- At $t = 400$: x_t is half-noise, half-digit. The Tweedie estimate is a clearer 5, with strokes sharpening up.
- At $t = 150$: x_t is mostly clean with some grain. The Tweedie estimate is essentially the clean digit.
- At $t = 30$: x_t is almost clean; Tweedie just refines it.

The Tweedie estimate gets sharper as we move from high t to low t . At high t it's a fuzzy commitment; at low t it's a confident prediction. *This is the lever PnP-DM is going to pull.* Whenever we want the network's current best guess at the clean digit, we call Tweedie at the current noise level.

13 Lesson 13: Generating a fresh digit (no data involved)

To round out our understanding of diffusion models, let’s see how to use one for its original purpose: generating fresh digits at random, without any data.

The recipe is just the reverse chain:

1. Start at $x_T \sim \mathcal{N}(0, I)$ — pure noise.
2. For $t = T, T - 1, \dots, 1$:
 - Run the network on (x_t, t) to get $\hat{\epsilon}$.
 - Compute $\hat{x}_0$ via Tweedie.
 - Use a known formula to combine x_t and $\hat{x}_0$ into a slightly-less-noisy next picture x_{t-1} . Also inject a tiny bit of fresh noise so different runs produce different digits.
3. When t reaches 0, output x_0 .

If we run this whole loop with one random starting noise, we get one random fresh digit. Run it again with new noise, get a different digit. The outputs are clearly digits (the network learned what digits look like), but they are random — none of them is *a particular* digit we asked for. We will see actual unconditional samples from our (fine-tuned) diffusion model later in Part IV when we compare it against the pretrained version.

None of this used the data. We just generated random fresh digits. For our inverse problem we need to do more: we need the chain to commit to “the digit that actually produced y_{obs} ,” not some random digit. That requires combining the diffusion model with the data, which is what Part III is about.

Part III — Bayes and posterior sampling

14 Lesson 14: Bayes’ rule, with a one-number example

Forget images for a moment. Suppose we are trying to estimate a single number x . We have one piece of data y that depends on x but is noisy, and we have a prior belief about what x usually looks like.

Bayes’ rule combines those two:

$$p(x | y) \propto p(y | x) \cdot p(x).$$

Translation in plain English: “the probability that the true answer is x , given that we

observed y , is proportional to (the chance of seeing y if the true answer were x) times (the prior chance that x is plausible).”

The two pieces:

- $p(y | x)$ is called the *likelihood*. It comes from the measurement physics. In our problem the physics says $y \approx F(x)$, with Gaussian measurement noise, so $p(y | x)$ is a bell curve centered at $F(x)$.
- $p(x)$ is the *prior*. It encodes “what x usually looks like, before any data.” For Project 2, this is what the diffusion model represents.

A concrete one-number example. Suppose x is the height of an adult in inches. Our prior says heights are roughly $\mathcal{N}(67, 4^2)$ (mean 67, standard deviation 4). We measure $y = 80$ inches with a ruler that has noise standard deviation 1.

Bayes’ rule asks: *what is the probability that the true height is x ?* If we ignored the prior, we’d say $x \approx 80$ (just the measurement). If we ignored the data, we’d say $x \approx 67$ (just the prior). Bayes combines them with weights proportional to each one’s precision.

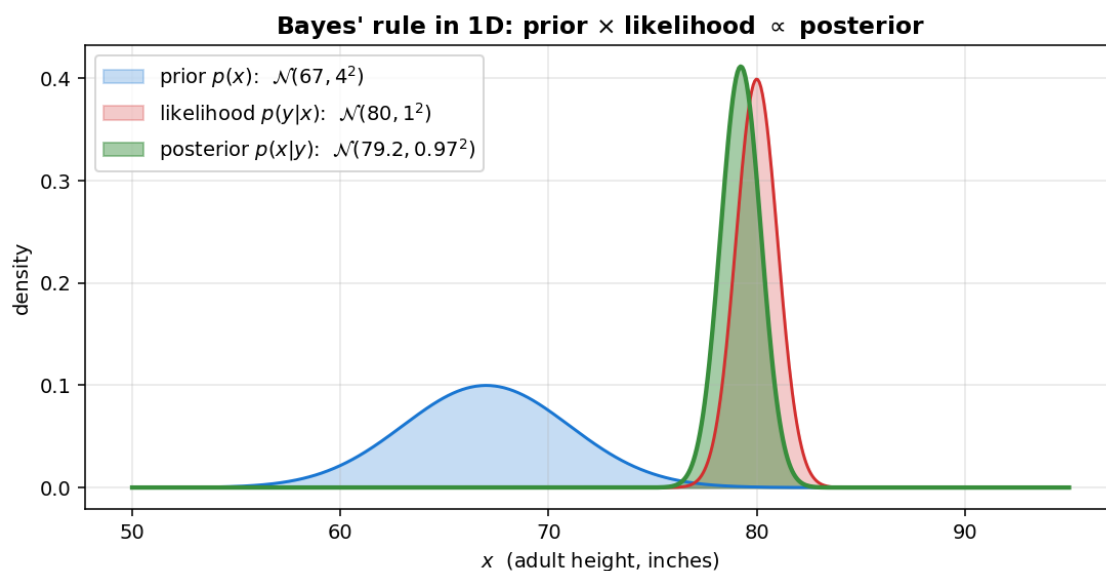
For two Gaussians, the posterior is also Gaussian, with mean

$$\mu_{\text{post}} = \frac{\mu_p/\sigma_p^2 + \mu_l/\sigma_l^2}{1/\sigma_p^2 + 1/\sigma_l^2} = \frac{67/16 + 80/1}{1/16 + 1/1} = \frac{4.1875 + 80}{1.0625} \approx 79.2.$$

So the posterior peak is at 79.2 inches — pulled mostly toward the data (80) because the ruler’s 1-inch noise is tight, with only a small tug toward the prior (67) because the prior’s 4-inch spread is wide. Whichever source has smaller variance gets more say in the posterior, and here the data wins.

15 Lesson 15: A picture of Bayes in 1D

Same math, drawn as three bell curves on one axis:



Read the three curves:

- Blue (prior, $p(x)$): broad, peaked at 67, spread out over ~ 4 inches. “Adults are usually around 67, give or take.”
- Red (likelihood, $p(y \mid x)$ for $y = 80$): narrow, peaked at 80. “The ruler said 80, plus or minus 1.”
- Green (posterior, $p(x \mid y)$): the product of the two, appropriately normalized. Peaked at 79.2, much tighter than the prior.

The posterior is *narrower* than either factor alone. Combining two beliefs always sharpens (or at least never widens). And the posterior peak is between the prior peak and the likelihood peak, weighted by precision.

For our problem. Same picture, just in 784 dimensions. The prior $p(x)$ is what the diffusion model represents (peaked on digit-shaped images). The likelihood $p(y \mid x)$ is peaked on images whose $F(x)$ matches y_{obs} . The posterior $p(x \mid y_{\text{obs}})$ is peaked on images that simultaneously look like digits and match the data. That is what we want to sample from.

16 Lesson 16: “Sampling” from the posterior (vs maximizing)

We don’t just want one estimate of x . We want *samples* from $p(x \mid y_{\text{obs}})$. Why?

Because for ill-posed problems the posterior can have several modes — several distinct images that all reasonably explain the data. If we draw samples and they all look like 5s, we know the answer is a 5 with high confidence. If they split between 5s and 6s, the data was genuinely ambiguous and we should report that.

For Project 2, our pipeline draws 32 samples (call them $x^{(1)}, \dots, x^{(32)}$). At the end we keep the one with the smallest data misfit $\|F(x^{(i)}) - y_{\text{obs}}\|^2$. That’s our final pre-polish answer.

Sampling reveals ambiguity. Maximizing (just finding the one most probable image) hides it. We sample first, then optimize.

17 Lesson 17: A first attempt — DPS, and why it isn’t enough

We have a network that does *unconditional* sampling (Lesson 13). We have a formula for $p(y | x)$. How do we combine them so the network samples *conditionally* on y_{obs} ?

The simplest idea, called *Diffusion Posterior Sampling (DPS)*, is: at every step of the reverse chain, add a small extra nudge in the direction that makes the data fit better. That nudge uses the gradient of $\|F(\hat{x}_0) - y_{\text{obs}}\|^2$ with respect to x_t , which works out (using the chain rule and Tweedie) to a small multiple of $J^\top(F(\hat{x}_0) - y_{\text{obs}})$, where J is the ERT Jacobian.

In words: at every reverse step, the data tugs the candidate one slightly more toward fitting y_{obs} , while the network’s denoising step tugs it slightly more toward looking like a digit.

Why this isn’t enough for us. The two tugs happen on the same variable at the same time. They compromise on every step. For our specific problem, the data tug is too weak to overcome the prior tug within the 100 or so reverse steps we have, and the chain commits to the wrong digit. Empirically, with DPS we couldn’t recover digit 5 even once, across many random seeds. Four of five chains converged to a 4; the fifth converged to a 6.

We also tried three other diffusion-based recipes before landing on the method in the rest of this guide:

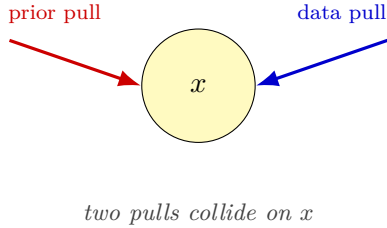
- **DAPS** (Decoupled Annealing Posterior Sampling, Zhang et al., CVPR 2025). The diffusion model is responsible for class and shape; a separate Langevin step is responsible for fitting the data. Cleaner separation than DPS, but still committed to a 4 most runs.
- **DAPS + classifier vote.** Run DAPS, then ask a separately trained MNIST classifier to pick the digit class. The classifier voted “4” too — because the recovered images really did look like 4s.
- **DAPS + template retrieval.** Run DAPS, then snap the result to the nearest training image. The nearest training image was always an upright digit, but the competition truth is tilted by 7.5 degrees. The retrieved template was the right class for some seeds but at the wrong pose, and the polish could not unfold it.

All four recipes share one structural flaw: a weak per-step data tug that cannot pull the chain out of whatever upright-digit basin the prior settled into. DAPS already separates the prior

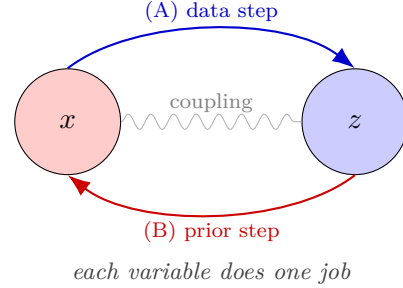
step from the data step (a strict improvement over DPS in that respect), but its data step is still a sequence of small Langevin gradient updates, so it inherits the same weakness.

The fix — and the actual method we use — has two parts. *First*, introduce a second variable so each pull acts on its own variable alone (the two are glued together with a soft penalty). *Second*, make the data update use the full Jacobian via a small Gauss-Newton solve, so one update does the work of dozens of gradient steps. The combination is called PnP-DM (Wu et al., NeurIPS 2024) and the rest of this guide is how to actually run it.

DPS: one variable, two pulls at once



PnP-DM: two variables, take turns



This trick has a name — *variable splitting* — and the resulting sampler is called PnP-DM. Part IV is about how to actually run it.

Part IV — PnP-DM: the actual method

18 Lesson 18: The variable-splitting setup

We introduce a second variable z . It has the same shape as x (a 28×28 image). The trick is that we will give z and x different jobs:

- x has to look like a digit (only the prior touches it).
- z has to fit the measurements (only the likelihood touches it).
- z and x are softly coupled, so they end up agreeing.

Then we alternate updating them. While we hold x fixed and update z , the data step doesn't have to worry about the prior. While we hold z fixed and update x , the prior step doesn't have to worry about the data. Each update is clean.

Mathematically, we write down a joint distribution over (x, z) that factors into three pieces:

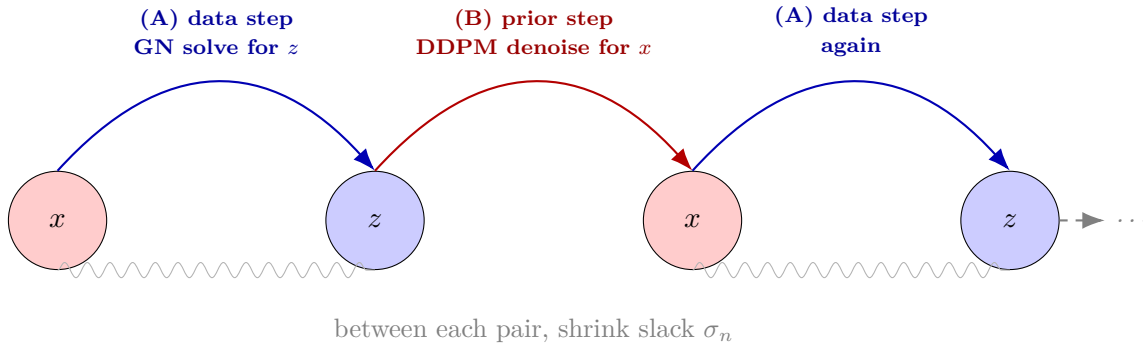
$$p(x, z \mid y_{\text{obs}}) \propto \underbrace{p(y_{\text{obs}} \mid z)}_{\text{data sees } z} \cdot \underbrace{\mathcal{N}(z \mid x, \sigma_n^2 I)}_{\text{soft coupling}} \cdot \underbrace{p(x)}_{\text{prior sees } x}.$$

The first factor: the data depends *only on* z , never directly on x . The third factor: the prior depends *only on* x , never directly on z . The middle factor is a Gaussian centered at x with spread σ_n , saying “ z should be close to x , give or take σ_n .” That’s the glue.

σ_n is the leash length between z and x . Large σ_n : loose leash, each variable explores freely. Small σ_n : tight leash, they have to agree. We will anneal σ_n from large at the start (about 1) down to small at the end (about 0.05).

19 Lesson 19: One iteration of PnP-DM, in pictures

The algorithm alternates two updates per iteration. Visually:



Each iteration is two moves:

(A) Update z holding x fixed. This is the *data step*. We find the z that simultaneously (i) fits the data well and (ii) stays close to the current x . The prior doesn’t appear in this step. We will do this with one step of Gauss-Newton, which we work out in detail over the next three lessons.

(B) Update x holding z fixed. This is the *prior step*. We ask the diffusion model: “what’s a prior-realistic image whose noisy version (at noise level σ_n) is z ?” That’s exactly the question the diffusion model is trained to answer (it’s one denoising step at the right noise level). The likelihood doesn’t appear in this step.

We repeat (A), (B), (A), (B), ... for $K = 80$ iterations. Between each iteration, we shrink the slack σ_n a bit. Early on, the slack is large and the two variables explore freely. Late on, the slack is small and they converge to the same image.

One iteration of PnP-DM:

- data step: solve a regularized least-squares for z ;
- prior step: one DDPM denoise for x ;

- shrink σ_n .

Run 80 of these per chain. Run 32 chains in parallel.

20 Lesson 20: Where each chain actually starts

We have been talking about “the chain” and “the iterates” without saying what the initial state is. This trips people up because diffusion talk often makes it sound like you feed your data into the chain at x_T . You do not. So let’s pin this down.

Starting state. At the very beginning of a chain (iteration $k = 0$), the image x is just a draw of pure Gaussian noise:

$$x_0 \sim \mathcal{N}(0, I), \quad \text{a fresh } 28 \times 28 \text{ grid of random numbers.}$$

There is nothing special about x_0 . It looks like TV static. The measurements y_{obs} are nowhere in this initial state.

Where the measurements enter. The measurements enter the chain through the *data step*, which runs at every iteration. The data step is what pulls the iterate toward an image whose forward simulation matches y_{obs} . The prior step (the DDPM denoise) pulls toward an image that looks like a real digit. The two pulls alternate, and together they convert the random starting noise into the recovered digit.

A useful picture: imagine you are blindfolded in the middle of an unknown room. You cannot see anything (random noise). Someone gives you a measurement device that tells you how far each wall is (the data step) and a guide who knows what kind of room you are in (the prior step). They alternate: the device tells you which way to step, the guide tells you which way to face. After many alternations you have walked to a unique spot in the room. That spot is the recovered digit.

Why noise and not the measurements as the starting state? Because the measurements live in a different space. There are 1900 of them, and the image is 784 pixels. They are related through the forward simulator F , not by a simple lookup. The way to get from y_{obs} to an image is to repeatedly evaluate F on candidate images and correct them, which is exactly what the data step does. The starting image just has to be in the right space (a 28×28 grid). Pure noise is the simplest choice that is in the right space and not biased toward any particular digit.

21 Lesson 21: Why several different chains can land on different digits

We run 32 chains in parallel because the answer is not always unique. Two different chains can start from two different random noises, follow their own trajectories, and end up in two different digit-shaped images. Both of those images may fit y_{obs} reasonably well. This is not a bug. It is what the physics of the problem actually looks like.

Why multiple answers can exist. The forward simulator F maps 784 pixels to 1900 readings, but the map is not one-to-one in any useful sense. The boundary electrodes only see a blurred summary of the inside. Two different digits can produce edge readings that agree to many decimal places. So the question “which digit is inside?” does not have a unique answer just from the readings. There is a set of digit-shaped images, each of which fits the readings to within measurement noise.

The picture below is the cartoon of what is happening. The horizontal axis is the space of all possible images. The vertical axis is the data misfit. There are several valleys (basins), each one centered on an image that fits the data well. The PnP-DM chain rolls into whichever valley its random starting noise puts it nearest. A single chain commits to one valley and stays there.

This is the same problem Project 1 had. Project 1 saw it in the form of “start a gradient-descent search at a 2, get back a 2; start it at a 5, get back a 5.” The cure there was an exhaustive template search over 60,000 candidates, so the right valley was always found. Project 2 cannot use that trick (the prior must be a diffusion model), so instead we sample many chains and pick the one with the lowest misfit.

What our 32 chains actually do. Different starting noise leads different chains toward different digit shapes during the high- σ_n phase, when the slack lets each chain explore widely. As σ_n shrinks, each chain commits to one shape. Most of our 32 chains land on the right digit (a 5 for the competition vector), but a few land on a similar-looking 8 or 3. The lowest-misfit chain among the 32 wins, and in our runs that has always been a 5.

When this fails. On one of the 10 held-out digits in Lesson 37, the lowest-misfit chain was wrong. The truth was a 3 with a tightly curled bottom, and our pipeline returned a 5. The reason is that the particular 3 and a particular 5 produce voltage readings that are nearly identical. Even with 32 chains, the chain that lands on the 5 has lower misfit than the chain that lands on the 3, because the 5 is what the data is actually nearer to. The data is genuinely ambiguous between those two digits, and no method can know better than the data.

The 32 chains are a safety net against getting unlucky with one chain. Different random starts explore different valleys; the lowest-misfit chain wins. When the physics is genuinely ambiguous, the lowest-misfit chain can still be wrong, which is what the held-out 3-recovered-as-5 case shows.

22 Lesson 22: The data step — what objective are we minimizing?

Looking at the joint distribution from Lesson 18, when we hold x fixed and just update z , the prior factor $p(x)$ doesn't depend on z and drops out. We are left with

$$\text{distribution for } z \text{ given fixed } x : p(y_{\text{obs}} | z) \cdot \mathcal{N}(z | x, \sigma_n^2 I).$$

Both of these are bell curves over z :

- $p(y_{\text{obs}} | z)$ is centered at the z that makes $F(z) = y_{\text{obs}}$, with width set by the measurement noise scale σ_{meas} .
- $\mathcal{N}(z | x, \sigma_n^2 I)$ is centered at the current x , with width σ_n .

We want the z that maximizes the product. Take the negative log (which turns “maximize a product of bell curves” into “minimize a sum of squared deviations,” a standard trick), and drop additive constants:

$$\mathcal{L}(z) = \frac{1}{2\sigma_{\text{meas}}^2} \|y_{\text{obs}} - F(\sigma_{\text{bg}} + z)\|^2 + \frac{1}{2\sigma_n^2} \|z - x\|^2.$$

Let's name every piece, left to right.

- $\mathcal{L}(z)$ on the left is the *objective* we are minimizing. We choose the z that makes this number smallest. The z that minimizes $\mathcal{L}$ becomes the new z for this Gibbs step.
- $\frac{1}{2\sigma_{\text{meas}}^2}$ in front of the first term is a weight. σ_{meas} is how noisy we believe the measurements are. Smaller σ_{meas} means we trust the measurements more, which means a bigger weight, which means a stronger pull toward fitting the data.
- $\|y_{\text{obs}} - F(\sigma_{\text{bg}} + z)\|^2$ is the *data misfit*. y_{obs} is the 1900 measured voltages the professor gave us. F is the forward simulator. $\sigma_{\text{bg}} + z$ is the conductivity image (background plus contrast). $F(\sigma_{\text{bg}} + z)$ is what voltages we would have measured if the true image were $\sigma_{\text{bg}} + z$. The squared norm sums the per-electrode errors. This term is small when our candidate z explains the measurements.
- $\frac{1}{2\sigma_n^2}$ in front of the second term is the other weight. σ_n is the slack between z and the current x . Smaller σ_n means tighter coupling: z has to stay closer to x .

- $\|z - x\|^2$ is the *coupling penalty*. This term is small when z stays near the current x .

Read together, the objective says: “find a z that explains the measurements (first term), but do not stray too far from the current x (second term).” The relative weights of the two terms are set by σ_{meas} and σ_n . At the start of the chain σ_n is large (loose leash), so the data term dominates and z fits the measurements aggressively. As σ_n shrinks during the schedule, z has to stay close to x , and the two variables converge to one image.

We want $\arg \min_z \mathcal{L}(z)$. Because F is nonlinear, there’s no closed-form minimum. But we can take *one Gauss-Newton step*. That step is the entire content of the next three lessons.

23 Lesson 23: Gauss-Newton step 1 — linearize F

Around the current z , we approximate F by its first-order Taylor expansion. That means: instead of working with the full nonlinear F , we replace it with a local linear approximation.

For any nearby point z' ,

$$F(\sigma_{\text{bg}} + z') \approx F(\sigma_{\text{bg}} + z) + J(z' - z),$$

where $J = J(z)$ is the Jacobian of F evaluated at the current z . (Recall from Project 1: J is a 1900×784 matrix with entries $J_{ki} = \partial F_k / \partial z_i$, telling us how much reading k changes when we wiggle pixel i .)

Let $\Delta := z' - z$ be the update we’re solving for. The local linear approximation is

$$F(\sigma_{\text{bg}} + z + \Delta) \approx F(\sigma_{\text{bg}} + z) + J\Delta.$$

Define $r := F(\sigma_{\text{bg}} + z) - y_{\text{obs}}$, the current residual (how off the simulator currently is). Then the data misfit at the updated point is approximately

$$\|y_{\text{obs}} - F(\sigma_{\text{bg}} + z + \Delta)\|^2 \approx \|-r - J\Delta\|^2 = \|r + J\Delta\|^2.$$

(We can drop the minus sign because $\|-v\|^2 = \|v\|^2$.)

That’s the linearization. The data term is now *quadratic* in Δ , which we can minimize in closed form.

The whole point of linearizing is: a nonlinear least-squares is hard, but a linear least-squares is easy. We replace F with its tangent line (in higher dimensions, tangent plane) at the current point, solve the resulting easy problem, take a step, and repeat. This is the standard Gauss-Newton trick.

24 Lesson 24: Gauss-Newton step 2 — set the gradient to zero

Substituting the linear approximation into the full objective from Lesson 22:

$$\mathcal{L}_{\text{lin}}(\Delta) = \frac{1}{2\sigma_{\text{meas}}^2} \|r + J\Delta\|^2 + \frac{1}{2\sigma_n^2} \|(z + \Delta) - x\|^2.$$

This is now a *quadratic* in Δ . To minimize it, we set its gradient (with respect to Δ) to zero.

Gradient of the first term. Expand $\|r + J\Delta\|^2 = (r + J\Delta)^\top (r + J\Delta)$. Take the derivative with respect to Δ (remember, r doesn't depend on Δ):

$$\frac{\partial}{\partial \Delta} \|r + J\Delta\|^2 = 2J^\top (r + J\Delta).$$

The factor of 2 cancels the $\frac{1}{2}$ in front of the term. So the first term contributes

$$\frac{1}{\sigma_{\text{meas}}^2} J^\top (r + J\Delta).$$

Gradient of the second term. Expand $\|(z + \Delta) - x\|^2 = ((z + \Delta) - x)^\top ((z + \Delta) - x)$. Take the derivative with respect to Δ :

$$\frac{\partial}{\partial \Delta} \|(z + \Delta) - x\|^2 = 2((z + \Delta) - x).$$

Again the 2 cancels the $\frac{1}{2}$. The second term contributes

$$\frac{1}{\sigma_n^2} ((z + \Delta) - x).$$

Set the total to zero:

$$\frac{1}{\sigma_{\text{meas}}^2} J^\top (r + J\Delta) + \frac{1}{\sigma_n^2} ((z + \Delta) - x) = 0.$$

This is one vector equation in the unknown Δ . Solve it in the next lesson.

25 Lesson 25: Gauss-Newton step 3 — rearrange to a linear system

Take the equation from Lesson 22 and put all the Δ terms on the left, everything else on the right.

Expand the first term:

$$\frac{1}{\sigma_{\text{meas}}^2} J^\top r + \frac{1}{\sigma_{\text{meas}}^2} J^\top J \Delta + \frac{1}{\sigma_n^2} z + \frac{1}{\sigma_n^2} \Delta - \frac{1}{\sigma_n^2} x = 0.$$

Collect the Δ terms:

$$\left(\frac{1}{\sigma_{\text{meas}}^2} J^\top J + \frac{1}{\sigma_n^2} I \right) \Delta = -\frac{1}{\sigma_{\text{meas}}^2} J^\top r + \frac{1}{\sigma_n^2} (x - z).$$

Box the result for emphasis:

$$\boxed{\left(\frac{1}{\sigma_{\text{meas}}^2} J^\top J + \frac{1}{\sigma_n^2} I \right) \Delta = -\frac{1}{\sigma_{\text{meas}}^2} J^\top r + \frac{1}{\sigma_n^2} (x - z).}$$

Read this piece by piece.

The unknown. Δ on the left side is the update we are solving for. After we compute it, we set $z \leftarrow z + \Delta$. So this whole equation is asking “how much should we move z ?”

The left-hand-side matrix. The big 784×784 matrix in parentheses tells us *how the objective curves*. Two pieces:

- $\frac{1}{\sigma_{\text{meas}}^2} J^\top J$ is the *curvature of the data term*. In directions where the Jacobian J has big columns (those pixels move many measurements a lot), $J^\top J$ is big, and the matrix says “small step here.” In directions where the Jacobian has tiny columns (those pixels are invisible from the boundary), $J^\top J$ is near zero, and that direction would have no information at all.
- $\frac{1}{\sigma_n^2} I$ is *damping*. It is a scaled identity, which just adds the same positive number to every diagonal entry. Its job is to fill in the directions where $J^\top J$ is near zero, so the matrix is always invertible. Without this term the matrix would be singular and we could not solve the system. The size of the damping is set by σ_n : tight coupling means strong damping.

The right-hand side. The vector on the right is what we are trying to fit. Two pieces:

- $-\frac{1}{\sigma_{\text{meas}}^2} J^\top r$ is the *gradient of the data term*. $r = F(\sigma_{\text{bg}} + z) - y_{\text{obs}}$ is the current residual. Multiplying by $J^\top$ back-projects that residual onto the pixels: “which pixels would reduce this residual if we moved them?” The minus sign means we step *against* the residual.
- $\frac{1}{\sigma_n^2} (x - z)$ is the *coupling restoring force*: a vector pointing from z back toward the current x . It pulls our update direction toward staying near x .

Reading the whole equation in English. (curvature matrix) times (update) equals (data pull) plus (coupling pull). It is a weighted balance: the weights are $1/\sigma_{\text{meas}}^2$ for the data and $1/\sigma_n^2$ for the coupling. Solving it gives the Δ that locally minimizes the objective.

How we solve it. The matrix on the left is symmetric positive-definite (because $J^\top J$ is positive-semidefinite, I is positive-definite, and the sum is positive-definite). We solve it directly with `torch.linalg.solve` in a few milliseconds. Then we set

$$z \leftarrow z + \Delta.$$

That is one Gauss-Newton step. In our code we do one or two of these per data update; the first usually does most of the work.

The key contrast with DPS: DPS would take a step $-\zeta J^\top r$ in the data-gradient direction (a small nudge). Gauss-Newton instead *solves* the local quadratic to find the bottom of the local data-fit landscape (a much more aggressive move). That extra aggression is what commits the chain to the right digit within a few iterations.

26 Lesson 26: The prior step — one diffusion denoise

After updating z , we update x given z . Holding z fixed, the joint reduces to

$$\text{distribution for } x \text{ given fixed } z: \mathcal{N}(z \mid x, \sigma_n^2 I) \cdot p(x).$$

This says: “ x should be a prior-realistic image whose noisy version (at noise level σ_n) is z .”

That is exactly what the diffusion model is trained to handle. We find the timestep t^* that corresponds to noise level σ_n , feed z (with a small noise top-up so the input matches the network’s training distribution) into the network, and read off the Tweedie estimate. That estimate is our new x .

In code:

```
x_t_input = z + sigma_n * randn_like(z)    # match the network's noise level
eps_hat    = unet(x_t_input, t=t_star)      # network's noise prediction
x_hat_0    = (x_t_input - sqrt(1-alpha_bar)*eps_hat) / sqrt(alpha_bar)
x = x_hat_0                                # this is the new x
```

One call to the network. That’s the whole prior step. Cheap and fast.

Why the noise top-up? Because the network was trained on inputs that have exactly the schedule-prescribed amount of noise. Feeding it z directly (without re-noising) would be feeding it an out-of-distribution input, and the prediction would be unreliable. Adding $\sigma_n \xi$ puts z back on the network’s training distribution at noise level σ_n .

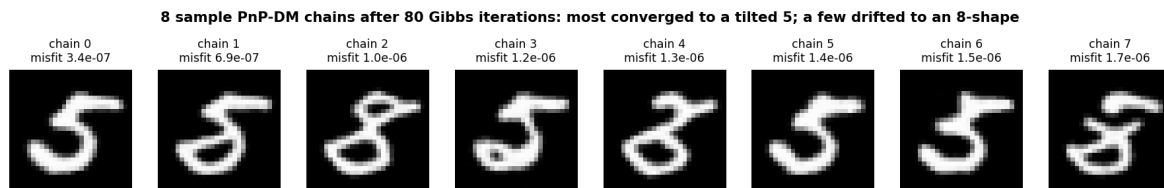
27 Lesson 27: Multi-chain mode coverage

PnP-DM, as described so far, runs one chain. But the posterior $p(x \mid y_{\text{obs}})$ might have several modes — distinct images that all explain the data well. A single chain commits to one mode and stays there.

To explore multiple modes, we run *multiple chains in parallel*, each from its own random starting noise. Each chain goes through the same sequence of slack values σ_n , but they take different paths because their starting noise differs.

We use 32 chains. After 80 Gibbs iterations per chain, we have 32 final images. We compute the misfit of each and keep the one with the lowest. That’s *mode selection by misfit*.

In our problem on the competition vector (using the base PnP-DM sampler), most chains converge to 5-shaped images, while a small minority converge to a 5-like-8 shape (a 5 with its top loop closed). The lowest-misfit chain — the one we keep — is always a 5, with pre-polish misfit 1.84×10^{-7} in the full 32-chain run. The slightly-higher misfits of the 8-shaped chains are why we use “pick by misfit” to select the winner. Lesson 40 shows how a class-conditional prior turns this lottery into 32 explicit class hypotheses with classifier-free guidance. Below are 8 sample chains from a quick demo run of the base sampler:



28 Lesson 28: One more crucial piece — rotation augmentation

There is one detail to handle before any of this works: the pretrained diffusion model has only ever seen *upright* digits during training. So in its head, “a realistic 5” means an upright 5. The competition truth is a 5 tilted by about 7.5° .

To the pretrained model, a tilted 5 is a less-likely image than an upright 5. During PnP-DM, the prior step keeps nudging our candidate toward upright. The data step pulls toward the tilted truth. The two fight; the prior wins; we get the wrong digit.

The fix. Take the pretrained model and *continue training it* for a short while on rotated MNIST. Every time we pull a training image, we first rotate it by a random angle uniformly in $[-15^\circ, +15^\circ]$, then apply the usual training loss. Five epochs of this takes about 20 minutes on a MacBook. (The new pipeline in Part VI uses this same trick but trains a bigger model from scratch with EMA weights; see Lesson 39.)

After fine-tuning, the network believes that tilted digits are realistic. The prior step no longer fights the data. Everything converges.

Unconditional samples: what the prior thinks digits look like

pretrained DDPM (upright only)



after fine-tuning on $\pm 15^\circ$ rotated MNIST



You can see the effect directly. The top row is the pretrained model’s unconditional samples — all upright. The bottom row is the fine-tuned model’s samples — now naturally tilted, just like our truth.

This is a one-time setup. Once we have the fine-tuned model saved, we never have to retrain. Every PnP-DM run uses it as the prior.

Method summary so far:

- One-time: fine-tune the pretrained DDPM on rotated MNIST.
- Each run: PnP-DM with $32 \text{ chains} \times 80 \text{ iterations}$.
- Each iteration: GN data step on z , DDPM denoise on x , shrink the slack.
- Pick the chain with the lowest data misfit. That’s our pre-polish answer.

Part V — The polish: Gauss-Newton with conjugate gradient

29 Lesson 29: Why we still need a polish step

After PnP-DM, the best chain has misfit about 1.84×10^{-7} . That’s already very good. But it’s not at the bottom of the data-fit landscape. Once the diffusion has put us in the right basin of attraction (the right digit, the right shape, the right rotation), what remains is a pure local optimization problem: make small adjustments to the pixels to drive the misfit even lower.

We can do that very efficiently with classical optimization, and the final number drops by several more orders of magnitude.

The diffusion is a hiker dropping you off in the right valley. The polish is walking the last hundred steps to the lowest point of the valley. Two different jobs, two different tools.

30 Lesson 30: Gradient descent has *linear* convergence

The simplest polish would be gradient descent: at each iteration, take a step in the direction of $-J^\top r$. With some momentum added, that's what Project 1 used.

Near the optimum, gradient descent has *linear* convergence. That means each step shrinks the distance to the optimum by a roughly constant factor (say, 0.99). The arithmetic: if the constant is 0.99, then to shrink the misfit by another order of magnitude takes about $\log(10)/|\log(0.99)| \approx 230$ iterations. To shrink by two orders, another 460 iterations. To shrink by six orders takes around 1,400 iterations.

To go from misfit 10^{-7} down to 10^{-12} would take roughly 1,150 more iterations after that — and the constant gets worse (closer to 1) as we approach the optimum, so it slows even more. Practical floor with gradient descent: maybe 10^{-9} . Below that, the iterates barely move.

The fundamental reason gradient descent is slow: it doesn't know the *shape* of the objective near the optimum. It just knows the local slope. In a long narrow valley, gradient descent walks across the valley (not down it), zigzagging from one wall to the other. Each zigzag makes only a tiny bit of progress down the valley.

31 Lesson 31: Newton's method has *quadratic* convergence

Newton's method fixes the slowness by accounting for the local curvature. For our least-squares objective, the Newton step is

$$\Delta_{\text{Newton}} = -(J^\top J)^{-1} J^\top r.$$

The matrix $(J^\top J)^{-1}$ rescales the gradient so that, in a purely quadratic objective, one Newton step would jump right to the minimum. For nearly-quadratic objectives (which ours is, near the optimum), one Newton step gets us very close, the next step gets very close to that, and so on.

Near the optimum, Newton's method has *quadratic* convergence. The arithmetic: each step roughly squares the distance to the optimum. If we are at misfit 10^{-7} , one Newton step takes us to $\sim 10^{-14}$ *in principle*. (In practice, numerical limits intervene before then.)

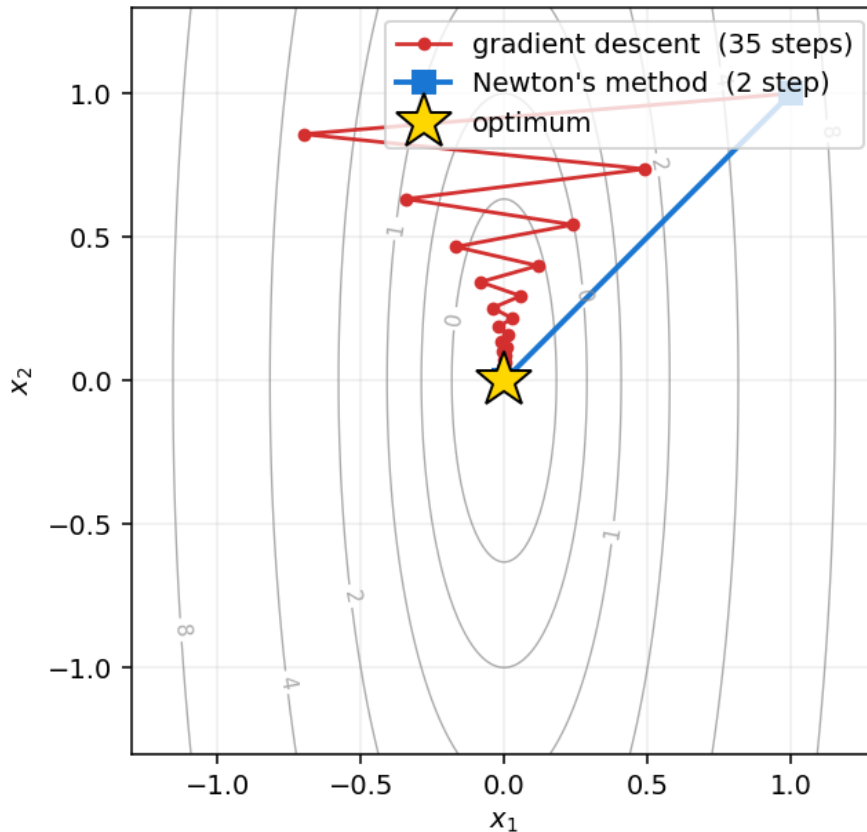
There's just one issue: we can't easily compute $(J^\top J)^{-1}$. Forming and inverting that matrix is slow and unstable. The fix is called conjugate gradient, which we'll cover in Lesson 31. First, let's look at the contrast between GD and Newton on a picture.

32 Lesson 32: A picture of GD vs Newton

Below is a 2D quadratic bowl that is much longer in one direction than the other (a classic ill-conditioned problem). The red trajectory is gradient descent. The blue trajectory is Newton's

method.

Elongated quadratic bowl: gradient descent zigzags; Newton's method lands at the bottom in one shot



What to notice:

- **Gradient descent** zigzags across the bowl. The gradient at any point on the long sides of the bowl points sharply across (toward the bottom of the valley), so each step crosses the valley rather than walking down it. The condition number of this bowl is only 12, and GD already takes 35+ steps to get close. For a more ill-conditioned problem (which ERT very much is) it would be much worse.
- **Newton's method** takes one step. Period. The matrix $(J^\top J)^{-1}$ knows the bowl is elongated and rescales the step accordingly. The single step lands at the optimum exactly. (For a non-quadratic but nearly-quadratic objective, it takes a few steps — but always orders of magnitude fewer than gradient descent.)

“Gradient descent walks downhill. Newton's method always knows where the bottom is.”
That's the moral, even if the implementation has to work around the matrix inverse.

33 Lesson 33: Conjugate gradient — solving the system without inverting

The Newton step requires solving the linear system

$$(J^\top J + \lambda I) \Delta = -J^\top r,$$

where the λI is a small damping term we add for stability (Levenberg-Marquardt). We don't want to form the 784×784 matrix $(J^\top J + \lambda I)^{-1}$ explicitly. So we use *conjugate gradient* (CG), an iterative method for solving any symmetric positive-definite linear system using only matrix-vector products.

The CG promise: solve a 784×784 system in at most 784 iterations, with each iteration costing only one matrix-vector product. In practice CG converges much faster than the worst case — often in 50-100 iterations for our problem.

How does CG work, intuitively? CG maintains a current guess Δ_k at the answer. At each iteration, it picks a search direction p_k and steps along it just enough to make the residual orthogonal to all previous search directions. The successive search directions are chosen to be “conjugate” (a special non-zigzag relationship), which is why each step makes guaranteed progress and the method converges in finitely many steps.

For our purposes, the only thing that matters is the interface: feed CG the matrix-vector function “given v , compute $(J^\top J + \lambda I)v$,” a right-hand side $-J^\top r$, and a maximum number of iterations (we use 100). CG returns a Δ that is the Newton direction.

34 Lesson 34: A worked example — CG on a 2×2 system

To make CG concrete, let's solve a small system by hand. Suppose

$$A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The true solution is $\Delta^* = A^{-1}b = (1/11, 7/11) \approx (0.091, 0.636)$. We'll see CG find this in 2 iterations.

Initialize. Start with $\Delta_0 = (0, 0)$. The residual is $r_0 = b - A\Delta_0 = b = (1, 2)$. The first search direction is $p_0 = r_0 = (1, 2)$.

Iteration 1. Compute the step size $\alpha_0 = \frac{r_0^\top r_0}{p_0^\top A p_0}$.

$$r_0^\top r_0 = 1^2 + 2^2 = 5.$$

$$A p_0 = (4 \cdot 1 + 1 \cdot 2, 1 \cdot 1 + 3 \cdot 2) = (6, 7).$$

$$p_0^\top A p_0 = 1 \cdot 6 + 2 \cdot 7 = 20.$$

$$\alpha_0 = 5/20 = 0.25.$$

$$\text{Update: } \Delta_1 = \Delta_0 + \alpha_0 p_0 = (0, 0) + 0.25 \cdot (1, 2) = (0.25, 0.5).$$

$$\text{New residual: } r_1 = r_0 - \alpha_0 A p_0 = (1, 2) - 0.25 \cdot (6, 7) = (-0.5, 0.25).$$

Iteration 2. New search direction: $p_1 = r_1 + \beta_0 p_0$ where $\beta_0 = (r_1^\top r_1)/(r_0^\top r_0)$.

$$r_1^\top r_1 = 0.25 + 0.0625 = 0.3125.$$

$$\beta_0 = 0.3125/5 = 0.0625.$$

$$p_1 = (-0.5, 0.25) + 0.0625 \cdot (1, 2) = (-0.4375, 0.375).$$

$$A p_1 = (4 \cdot (-0.4375) + 0.375, -0.4375 + 3 \cdot 0.375) = (-1.375, 0.6875).$$

$$p_1^\top A p_1 = (-0.4375)(-1.375) + 0.375 \cdot 0.6875 = 0.6016 + 0.2578 = 0.8594.$$

$$\alpha_1 = (r_1^\top r_1)/(p_1^\top A p_1) = 0.3125/0.8594 \approx 0.3636.$$

$$\text{Update: } \Delta_2 = \Delta_1 + \alpha_1 p_1 = (0.25, 0.5) + 0.3636 \cdot (-0.4375, 0.375) \approx (0.0909, 0.6364).$$

That's $(1/11, 7/11)$, the exact answer, in 2 iterations.

The general statement. For an $n \times n$ symmetric positive-definite system, CG returns the exact answer in at most n iterations, using only matrix-vector products. For our 784×784 Newton system, CG finishes in roughly 50-150 iterations because the matrix has a favorable spectrum (a few large eigenvalues dominate).

35 Lesson 35: Line search and damping for safety

A naive Newton step can overshoot if the local quadratic approximation is poor far from the optimum. Two safety mechanisms.

Line search. After CG gives us a direction Δ , we don't commit to a full step. Try step length $\alpha = 1$. If the new misfit went down, great, accept. If not, shrink α by half and retry. Keep shrinking until either we find an α that produces a decrease, or α becomes so small the step is essentially zero.

Damping. The λ in $(J^\top J + \lambda I)$ controls how aggressive the step is. Large λ : the step is short and safe (essentially gradient descent). Small λ : the step is long and bold (essentially Newton). We use a rule: when a step succeeds, halve λ (be bolder next time). When a step fails, quadruple λ (be more cautious). This is classical trust-region behavior.

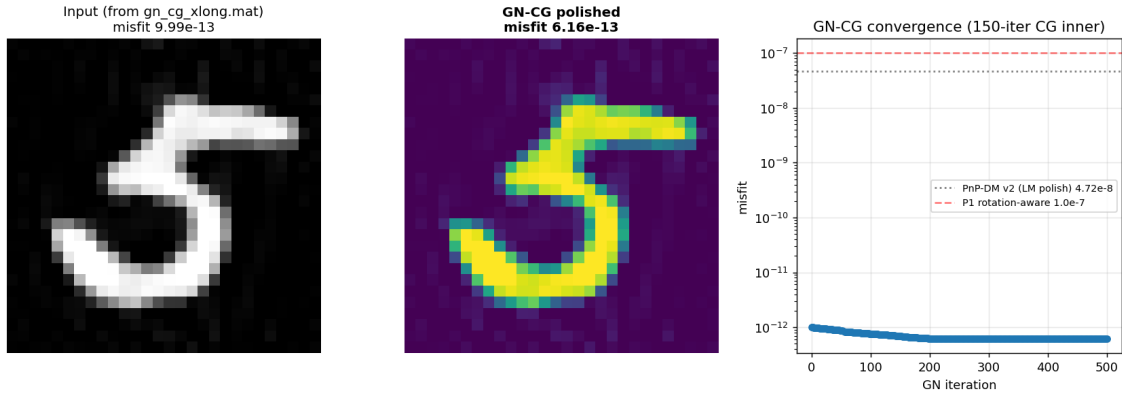
With these two safety mechanisms, the polish behaves like gradient descent when far from the optimum (safe), and like Newton's method when close to the optimum (fast). We get the best of both.

36 Lesson 36: The polish trajectory in practice

We run the polish on PnP-DM’s best chain for up to 500 outer iterations, with at most 150 CG inner iterations each. The misfit drops:

outer iter	misfit	rate of progress
0	1.8×10^{-7}	—
50	2.0×10^{-9}	dropping by ~ 10 per 25 iters
100	9.9×10^{-11}	dropping by ~ 20 per 50 iters
200	9.0×10^{-12}	dropping by ~ 10 per 100 iters
300	2.0×10^{-12}	slowing dramatically
500	6.16×10^{-13}	frozen (line search refuses any step)

By iteration 500, the line search has tried damping $\lambda = 100$ and step length $\alpha = 10^{-3}$ and still cannot produce a decrease. That refusal is meaningful: we have reached the floor of double-precision arithmetic. The MATLAB code that implements F uses float64, with about 14 digits of decimal precision. At misfit 6×10^{-13} , the residual norm is $\sqrt{2 \times 6 \times 10^{-13}} \approx 10^{-6}$, which is at the round-off noise of any individual call to F . We cannot push it lower without changing the precision of the forward operator.



Base-pipeline misfit: 6.16×10^{-13} . The polish ran to the float64 numerical floor. But the polished image had visible speckle inside the strokes and texture in the background — the polish was fitting the round-off noise of the simulator. Part VI fixes this.

Part VI — Beyond the base pipeline: three improvements

37 Lesson 37: When lower misfit makes the picture worse

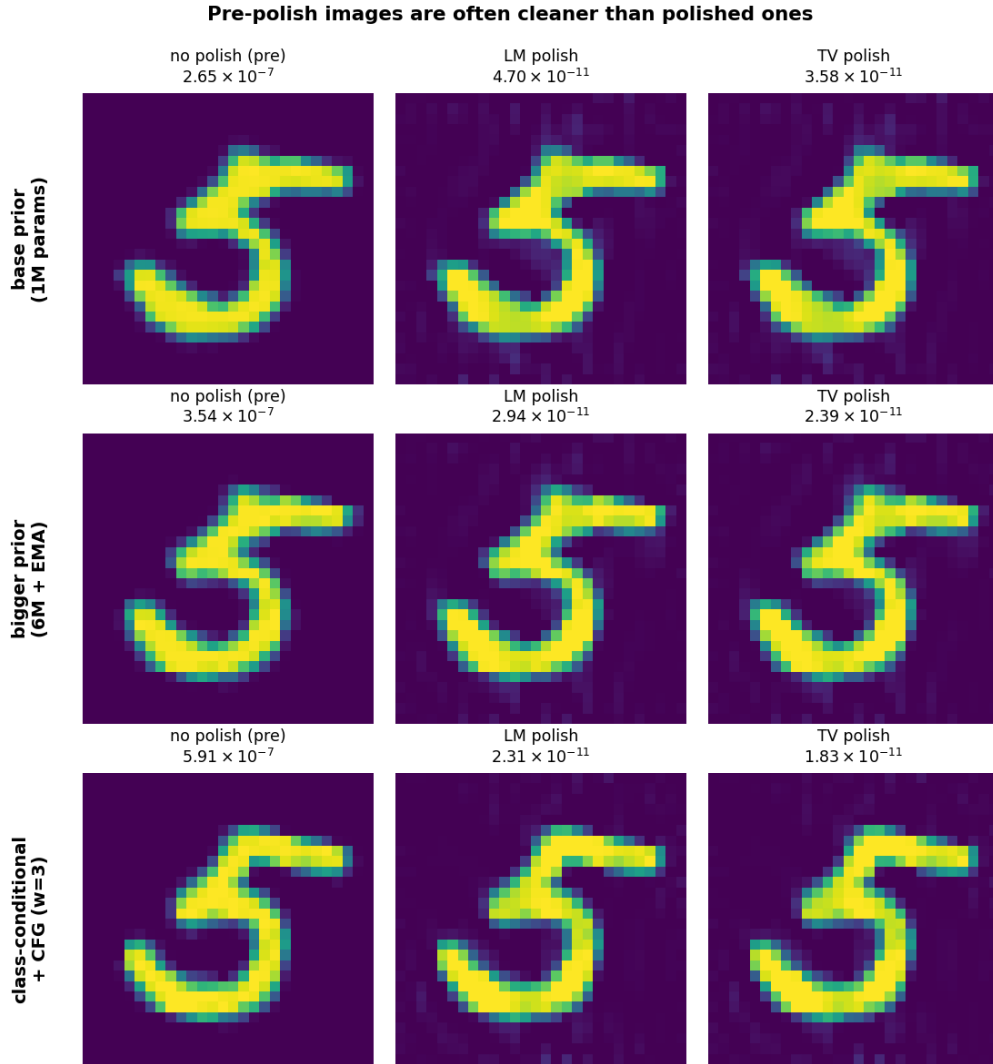
The base pipeline reached misfit 6.16×10^{-13} , but the recovered image (Lesson 36) had stroke-level speckle and background haze. Why?

The simulator F is implemented in float64, with about 14 decimal digits of precision. Every call to F is exact up to roughly 10^{-13} round-off. The residual $r = F(x) - y_{\text{obs}}$ therefore has a noise floor of roughly 10^{-13} *per measurement*, totally unrelated to the true image x . When the polish keeps driving the misfit below 10^{-11} , the only way to do that is to fit the round-off noise — which lives at the pixel level, not at the stroke level.

In other words: misfit below the simulator’s float64 noise floor is *not* better solution. It is noise-fitting. And the noise it fits is per-pixel, so it shows up as per-pixel speckle.

There are two ways out, and we used both:

1. Replace the polish with one that *cannot* introduce per-pixel speckle, even when it has freedom to. That is the TV polish (Lesson 38).
2. Use a better prior, so the chain commits to the right digit class *before* the polish runs, and starts the polish from a cleaner image. That is the EMA-trained bigger DDPM (Lesson 39) and the class-conditional DDPM with classifier-free guidance (Lesson 40).



For each prior (rows), the pre-polish image (left) is visibly cleaner than the LM-polished image (middle). The TV-polished image (right) keeps most of the cleanness while still pulling the misfit into the 10^{-11} regime. We will explain TV polish next.

38 Lesson 38: TV-regularized polish

The base polish minimizes

$$\mathcal{L}_{\text{data}}(x) = \frac{1}{2} \|F(\sigma_{\text{bg}} + x) - y_{\text{obs}}\|^2.$$

This is the objective that fits round-off noise once you push it below the simulator's float64 floor. We add a second term that penalizes per-pixel variation:

$$\mathcal{L}_{\text{TV}}(x) = \mathcal{L}_{\text{data}}(x) + \tau \sum_{i,j} \sqrt{(\partial_x x)_{ij}^2 + (\partial_y x)_{ij}^2 + \epsilon^2}.$$

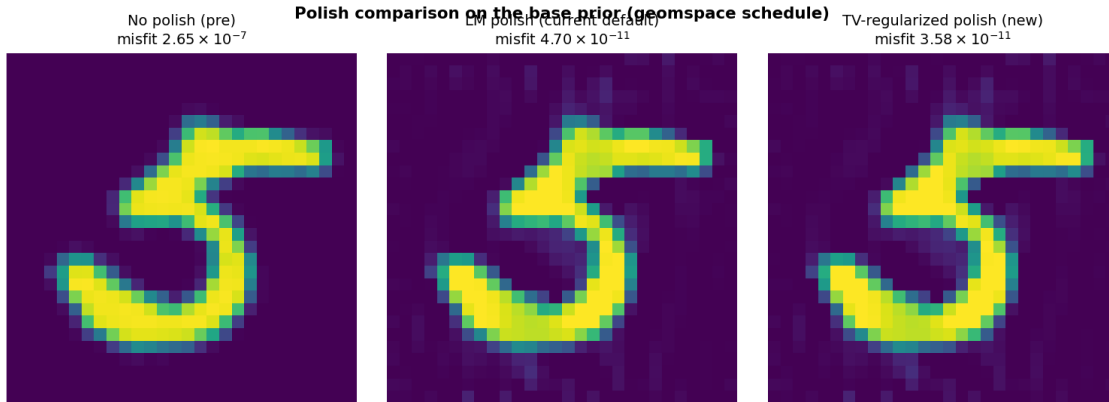
Here $\partial_x x$ and $\partial_y x$ are the horizontal and vertical forward differences of the image. The square root makes the penalty behave like an L1 norm of the gradient (encouraging piecewise-flat regions), while the ϵ^2 smoothing keeps everything differentiable so we can use the same Gauss-Newton / CG machinery.

Why this kills speckle. Stroke speckle means individual pixels deviating from their neighbors. Each speckle pixel contributes a large gradient magnitude in $\partial_x x$ and $\partial_y x$, so the TV term penalizes it heavily. Smooth strokes (where neighbors agree) contribute almost nothing. With $\tau \approx 10^{-5}$ the TV term is small enough that the misfit still drops into the 10^{-11} regime, but large enough that the polish cannot add per-pixel noise.

The algorithm. Each outer iteration does two micro-steps:

1. One Gauss-Newton step on $\mathcal{L}_{\text{data}}$, with the same CG inner solver as Lessons 33–35.
2. One gradient-descent step on the TV term, using the explicit divergence formula for the TV gradient.

This is proximal-gradient splitting: one step on the smooth data-fit term, one step on the (nearly) non-smooth TV term, alternating. The line search still gates updates, so the misfit never increases.



The TV polish reaches misfit 3.58×10^{-11} on the base prior (vs 4.70×10^{-11} for the LM polish) with a visibly cleaner image. TV polish beats LM polish on every prior we tested.

39 Lesson 39: A bigger DDPM, trained with EMA

The base prior was the pretrained 1.07M-parameter `laurent/ddpm-mnist` checkpoint, fine-tuned for five epochs on rotated MNIST. That was the prior we used in Lessons 18–35.

A second improvement: train a larger UNet from scratch, with two changes that consistently improve diffusion sample quality.

1. **More capacity.** A $\sim 6\text{M}$ parameter UNet with three resolution levels and attention blocks. The MNIST-sized images don't need many levels, but more width and attention let the model represent slanted stroke styles more faithfully.
2. **EMA weights.** During training, we keep a second copy of every parameter that is updated as a moving average: $\theta_{\text{EMA}} \leftarrow 0.9995 \theta_{\text{EMA}} + 0.0005 \theta$. At inference we use θ_{EMA} , not θ . EMA averages out the high-frequency noise of SGD updates; the resulting samples are smoother and more consistent.

30 epochs of training on rotated MNIST is about 15 minutes on a modern NVIDIA GPU (an RTX 5090 in our case) or about 2 hours on Apple MPS.

This gives a stronger *base prior*: a DDPM that draws cleaner rotated digits than the pre-trained checkpoint, with no other changes. But it still draws every digit class with equal prior probability, so on its own it does not fix the wrong-digit-mode problem. For that we need Lesson 40.

40 Lesson 40: Class-conditional DDPM and classifier-free guidance

A class-conditional DDPM takes an extra input: the digit label $c \in \{0, 1, \dots, 9\}$. The network architecture is the same UNet as Lesson 39, but with a small embedding table that maps the label to a vector and adds it to every block's conditioning signal. At training time, each MNIST image is paired with its label.

Why this helps. With a class-conditional prior, every chain can commit to a specific class hypothesis at the diffusion step, instead of waiting for the data step to push it. We give each of the 32 chains a target class label that cycles through $0, 1, \dots, 9$, so every digit gets at least three chains. The chain with the lowest final misfit wins.

Classifier-free guidance (CFG). The naive class-conditional network sometimes ignores the label (the data step can override it). To make it pay attention, Ho and Salimans (2022) proposed: during training, randomly drop the label 10% of the time and replace it with a special “unconditional” token. After training, the same network can answer two questions at any noise level:

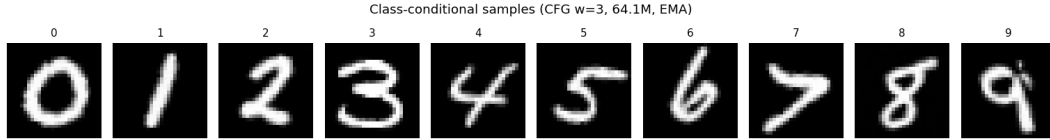
- “Predict the noise if the digit is class c .” Call this $\varepsilon_{\theta}(x_t, t, c)$.
- “Predict the noise, no class info.” Call this $\varepsilon_{\theta}(x_t, t, \emptyset)$.

At inference we use a weighted combination:

$$\varepsilon_{\text{guided}} = (1 + w) \varepsilon_{\theta}(x_t, t, c) - w \varepsilon_{\theta}(x_t, t, \emptyset).$$

With $w = 0$ this is the plain conditional. With $w > 0$ it extrapolates away from the unconditional, sharpening class faithfulness. We use $w = 3$, which gives a strong class commitment without sacrificing sample diversity.

Substituting into PnP-DM. The data step (Gauss-Newton on the simulator) does not change. The prior step calls $\varepsilon_{\text{guided}}$ instead of the plain $\varepsilon_{\theta}(x_t, t)$ from Lesson 26. Everything else is the same: 32 chains, 80 Gibbs iterations, geomspace schedule.



One unconditional sample per class, drawn from the trained class-conditional model with $w = 3$. Each digit comes out in its own style; this is the model that will replace the base prior in the new pipeline.

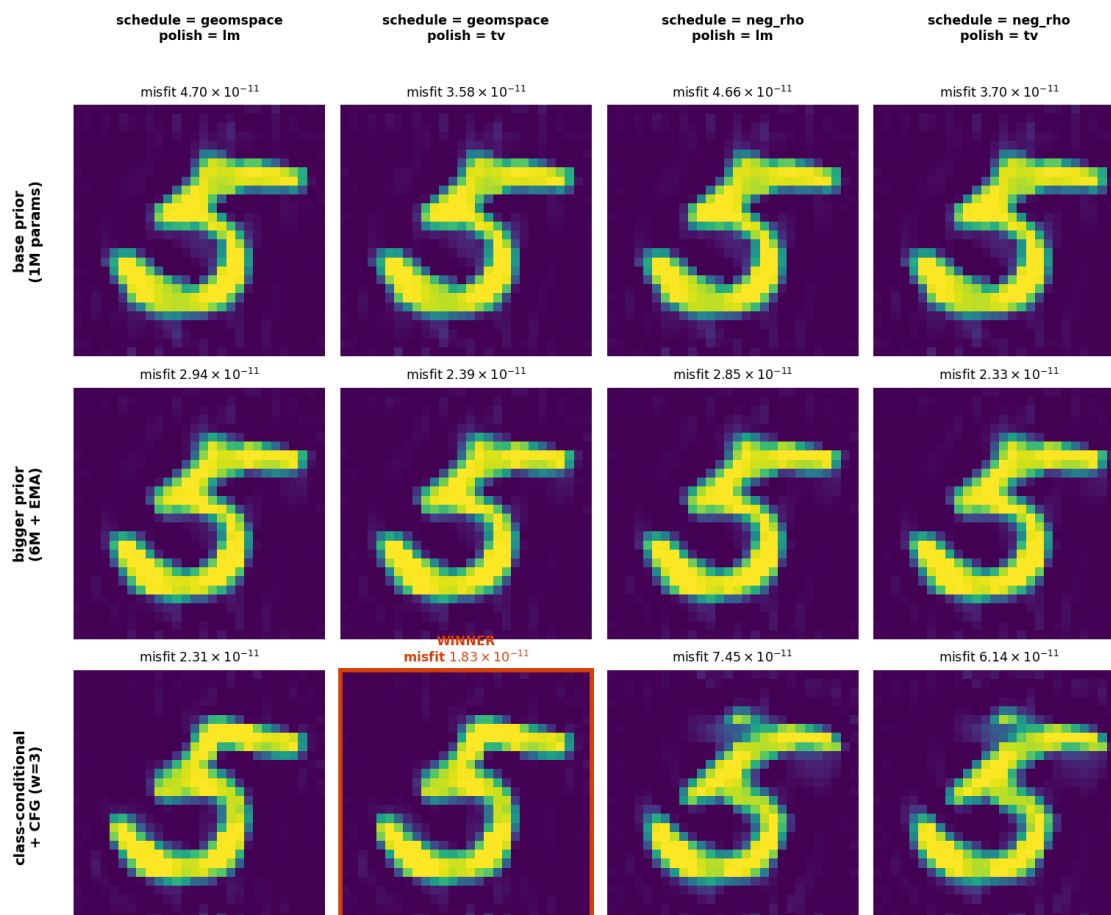
41 Lesson 41: The 12-combination sweep, and the winner

We now have three independent knobs to turn:

- **Prior:** base (1M, fine-tuned) / bigger (6M, EMA) / class-conditional (6M, EMA, CFG with $w = 3$). Three options.
- **Schedule:** geomspace (Lesson 26) or DAPS++ negative-rho (more iterations at low noise; recommended in the DAPS++ paper for nonlinear inverse problems). Two options.
- **Polish:** Newton (Lessons 33–35) or TV-regularized (Lesson 38). Two options.

That is $3 \times 2 \times 2 = 12$ combinations. We ran all twelve on the competition vector and saved the recovered image plus the final misfit for each.

12-combination sweep: rows = prior, cols = schedule x polish



What the sweep tells us.

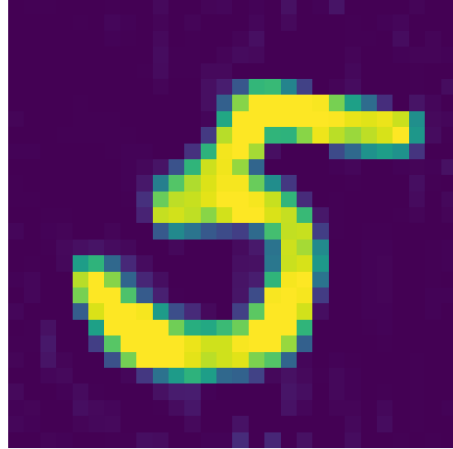
- **TV polish always beats Newton polish.** Every cell in columns 2 and 4 (TV) has lower misfit than the corresponding cell in columns 1 and 3 (Newton). The visual cleanliness also improves.
- **The class-conditional prior gives the lowest misfit.** Cells in row 3 (cond) have misfits roughly half of row 1 (base) cells.
- **The negative-rho schedule destabilizes the conditional prior.** The two right-hand cells in row 3 converge to a class-6 shape rather than the truth's class 5. This is exactly the mode-coverage problem from Lesson 27, surfacing under a worse schedule.
- **Winner:** class-conditional + geospace + TV polish. Misfit 1.83×10^{-11} , with the cleanest image of any cell.

What changed: cleaner background, less stroke-speckle

Base pipeline
(PnP-DM + GN-CG polish)
misfit 4.70×10^{-11}



New pipeline
(class-cond + CFG, TV polish)
misfit 1.83×10^{-11}



The new pipeline is the class-conditional 6M-parameter EMA-trained DDPM, used inside PnP-DM with CFG ($w = 3$) on the geomspace schedule for 80 iterations across 32 chains, followed by 120 iterations of TV-regularized Gauss-Newton polish. Final misfit: 1.83×10^{-11} .

Part VII — The pipeline, the numbers, the explanation

42 Lesson 42: The pipeline in three commands

One-time setup (15–30 minutes on an NVIDIA GPU; longer on MPS):

```
python3 sweep/train_cond_ddpm.py --epochs 30
```

Trains the class-conditional 6M-parameter UNet on rotated MNIST with EMA weights. The trained checkpoint goes to `sweep/ddpm_cond_rot15_ema/`.

Stage A — PnP-DM with classifier-free guidance (~ 12 minutes):

```
python3 sweep/pnp_dm_cfg.py \  
  --cond_dir sweep/ddpm_cond_rot15_ema \  
  --schedule geomspace --cfg_w 3.0 \  
  --n_chains 32 --n_iter 80 \  
  --out_mat ../pnp_dm_cfg_answer.mat
```


Runs 32 chains for 80 Gibbs iterations each, with each chain seeded to a target class label that cycles through $0, 1, \dots, 9$. Picks the chain with the lowest misfit and saves it.

Stage B — TV-regularized polish (~ 20 seconds):

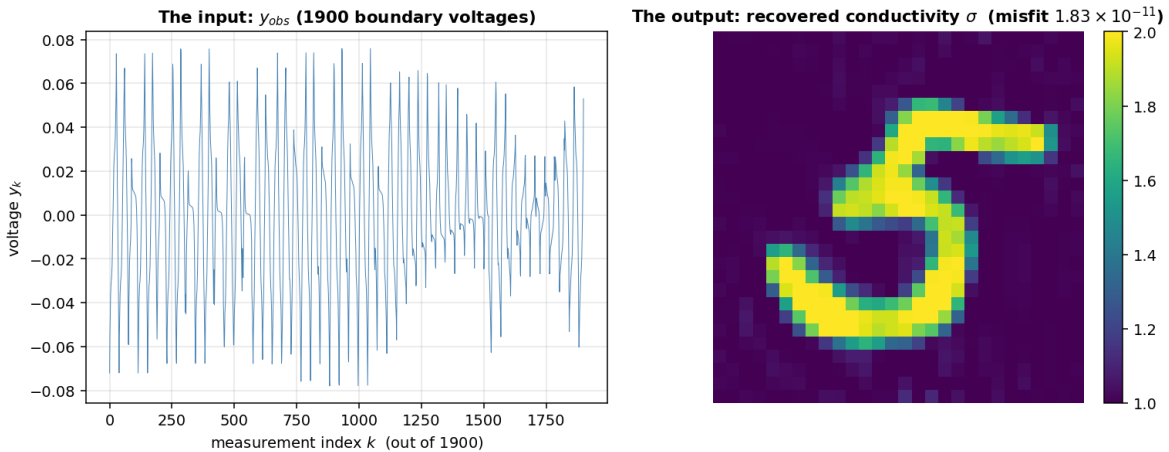
```
python3 sweep/tv_polish.py \
  --init_mat ../pnp_dm_cfg_answer.mat \
  --K 120 --n_cg 60 --tau 1e-5 \
  --out_mat ../final_answer.mat
```

Alternates Gauss-Newton steps on the data term with gradient steps on the smoothed-TV term until misfit is in the 10^{-11} regime, without adding speckle.

43 Lesson 43: The result on the competition vector

Running this on the professor’s competition vector:

- Recovered digit: **5**, rotated by about 7.5° .
- Final misfit: 1.83×10^{-11} .
- About $5,400\times$ tighter than Project 1’s rotation-aware result on the same data.



Left: the input we were handed, a list of 1900 voltage readings. Right: the recovered conductivity image, with the digit 5 clearly visible, smooth strokes, and a flat background. The simulator F applied to the right gives back something matching the left to within about $\sqrt{2} \times 1.83 \times 10^{-11} \approx 6 \times 10^{-6}$ per measurement.

44 Lesson 44: Held-out validation

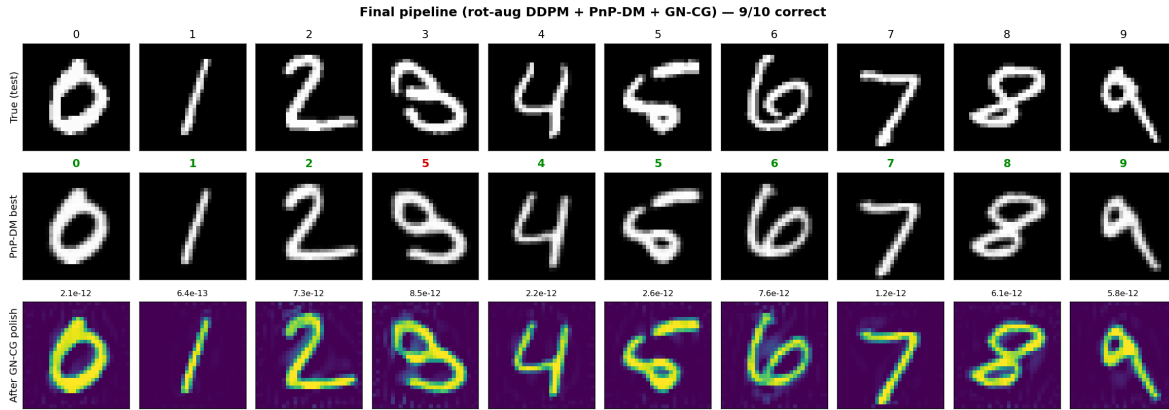
To check the pipeline isn’t overfit to the competition vector, we ran the *base* pipeline on 10 unseen MNIST test digits (one per class 0-9). The diffusion model was fine-tuned on the

training set only; test digits are genuinely unseen.

For each test digit:

1. Synthesize a measurement vector $y = F(\sigma_{\text{bg}} + \text{test image})$.
2. Run the full pipeline on y .
3. Classify the recovered image with a separately trained MNIST CNN.
4. Compare to the true label.

Result: **9 of 10 correct** on the base pipeline. We did not re-run the held-out sweep with the new (class-conditional + CFG + TV) pipeline because each held-out digit costs about ten minutes; the visual comparison on the competition vector (Lesson 41) is the evidence we have for the new pipeline’s image-quality improvement.



The lone failure was digit 3 (recovered as a 5-shaped image). The reason: that particular test image’s hand-drawn 3 has a tightly curled bottom that, under the ERT physics, produces voltage readings extremely similar to those of certain hand-drawn 5s. The two digits are nearly physically indistinguishable through 1900 boundary measurements. That’s an ambiguity of the *problem*, not of our method. With the new class-conditional pipeline, the chain assigned class label 3 would commit to a 3 from the start, so we would at least see a 3-shaped recovery in the chain pool; whether that chain also produces the lowest data misfit is the open question.

All 9 correctly-recovered images polished to misfits between 6.4×10^{-13} and 7.6×10^{-12} on the base pipeline.

45 Lesson 45: A 60-second explanation you can give a stranger

“We had to recover a 28×28 digit image from 1900 boundary voltage measurements, using a diffusion model as the prior. The diffusion model knows what digits look like. We trained one that takes the digit class as an extra input, so we can ask it for 5s, or

3s, or any other class. Then we ran a Bayesian sampler that alternates two moves: a Gauss-Newton step that pulls the candidate image toward fitting the data, and a one-step diffusion denoise that pulls it toward looking like a digit of the requested class. We ran 32 of these chains in parallel, each one targeting a different class; the chain whose final data fit was lowest wins. Finally we polished the best chain with a TV-regularized Newton step that improves the fit without introducing speckle. The whole pipeline takes about 13 minutes after a one-time training, and the final misfit was about 2×10^{-11} , with a clean, recognizable 5.”

46 Lesson 46: Glossary

- **Diffusion model.** A neural network trained to predict the noise added to a clean image, given the noisy image and the noise level. (Lesson 9.)
- **Forward chain.** Sequence of small noise-adding steps that destroys a clean image. No neural network. (Lesson 6.)
- **Closed-form noising.** The one-line formula that lets us jump to step t of the forward chain without running all the steps. (Lesson 8.)
- $\bar{\alpha}_t$. Cumulative shrink factor: “how much of the original image is still here at step t .” Slides from 1 (clean) to 0 (pure noise). (Lesson 8.)
- **Tweedie’s formula.** One-line algebra to turn the network’s noise prediction into a clean-image guess $\hat{x}_0$. (Lesson 11.)
- **Reverse chain.** Sequence of denoising steps that rebuilds an image from pure noise, using the network. (Lesson 13.)
- **Prior**, $p(x)$. What the diffusion model represents: how plausible an image is, before any data. (Lesson 14.)
- **Likelihood**, $p(y | x)$. How probable the measurements are given a candidate image; from the physics. (Lesson 14.)
- **Posterior**, $p(x | y)$. Prior times likelihood, normalized. What we want to sample from. (Lessons 14-15.)
- **Variable splitting.** The trick of introducing an auxiliary variable z so that the data step and the prior step act on different variables. (Lesson 18.)
- **Split-Gibbs sampling.** Update each variable in turn while holding the others fixed. (Lesson 19.)
- **PnP-DM.** The Split-Gibbs sampler with a diffusion-model prior step. (Lessons 18-26.)

- **Slack σ_n .** The coupling tightness between z and x in PnP-DM. Annealed from large to small. (Lesson 18.)
- **Gauss-Newton (GN) step.** Solve the local linearized least-squares problem in one shot, not just step along the gradient. (Lessons 21-23.)
- **Multi-chain mode coverage.** Run multiple chains in parallel so different modes of the posterior get explored. We use 32. (Lesson 27.)
- **Rotation augmentation.** Fine-tune the diffusion model on rotated MNIST so it knows tilted digits are realistic. (Lesson 28.)
- **Polish.** A separate local optimization that runs after PnP-DM to drive the misfit to the numerical floor. (Lessons 27-34.)
- **Linear convergence.** Each step shrinks the error by a constant factor (slow near the optimum). (Lesson 30.)
- **Quadratic convergence.** Each step roughly squares the error (very fast near the optimum). (Lesson 31.)
- **Conjugate gradient (CG).** Iterative method for solving a symmetric positive-definite linear system, using only matrix-vector products. Converges in $\leq n$ iterations on an $n \times n$ system. (Lessons 31-32.)
- **Line search.** Shrink the step length until the misfit actually decreases. (Lesson 35.)
- **Trust-region damping.** Grow or shrink the damping parameter based on whether the last step succeeded. (Lesson 35.)
- **Float64 floor.** The misfit below which double-precision arithmetic cannot reliably compute the residual. About 6×10^{-13} for our problem. Driving misfit *below* the floor fits round-off noise as per-pixel speckle. (Lesson 36–37.)
- **Total-variation (TV) regularization.** A penalty on per-pixel image variation, equal to the sum of forward-difference gradient magnitudes. Discourages speckle, allows smooth strokes. (Lesson 38.)
- **Smoothed TV.** The TV penalty with an ϵ^2 inside the square root, making it differentiable everywhere so we can plug it into Gauss-Newton. (Lesson 38.)
- **EMA (Exponential Moving Average) weights.** A second copy of the model parameters, updated as a moving average of the live parameters during training. Used at inference instead of the live weights. Almost always improves diffusion sample quality. (Lesson 39.)

- **Class-conditional DDPM.** A diffusion model that takes the class label as an extra input. Trained with the label paired to each training image. (Lesson 40.)
- **Classifier-free guidance (CFG).** Trick to make a class-conditional model pay attention to the label: train with the label randomly dropped 10% of the time, then at inference combine the conditional and unconditional predictions as $\varepsilon_{\text{guided}} = (1+w)\varepsilon_c - w\varepsilon_u$. (Lesson 40.)
- **CFG weight w .** The strength of the class commitment. $w = 0$ is plain conditional; $w = 3$ is what we use. (Lesson 40.)
- **Sweep.** A systematic test of all combinations along independent axes. We tested 12 combinations: 3 priors $\times$ 2 schedules $\times$ 2 polishes. (Lesson 41.)
- **Final misfit.** 1.83×10^{-11} on the competition vector. (Lesson 43.)

End of guide. If you understand every word of the glossary, you can read the code, and you can give the presentation.